		experience the greater magnetic flux density and hence change in magnetic flux linkage.
27	**D**	The I_{rms} of the square waveform is I_o.

		(working: $I_{rms} = \sqrt{\dfrac{I_o^2 \times \frac{T}{2} + I_o^2 \times \frac{T}{2}}{T}} = I_o$) The average power is given by $P = I^2 R = (I_o)^2 R$ When the current is doubled to $2\ I_o$, the new I_{rms} is $2\ I_o$. (similar working to above) The new average power is given by $W = I^2 R = (2\ I_o)^2 R$ $= 4\ (I_o)^2 R = 4\ P$.
28	**D**	To find the p.d. V_p in the primary coil,